



भारतीय सूचना प्रौद्योगिकी संस्थान गुवाहाटी
INDIAN INSTITUTE OF INFORMATION TECHNOLOGY GUWAHATI

CS351: IT Workshop II (Cloud Computing), Mid Semester Examination, Monsoon 2024

Note: Maximum time: 2 Hours. Total Marks: 40

- Answer all of the following questions. Answers must be brief and to the point.
- Make reasonable assumptions wherever necessary and state them.

- ✓ (a) [2 points] In cloud computing, multiple customers can share a cloud resource while maintaining data isolation. Justify how.
✓ (b) [3 points] Suppose a cloud-based application is experiencing a spike in traffic. Explain how cloud computing can handle this situation.
- ✓ (a) [2 points] Differentiate between a type-1 hypervisor and a type-2 hypervisor.
✓ (b) [3 points] Does the guest operating system need to be modified to run in a virtual machine? Justify
- ✓ (a) [2 points] Can a load balancer ensure high availability in cloud computing? Explain.
✓ (b) [3 points] Monitoring in cloud computing can help to ensure that the service level agreement is met. Justify how.
- ✓ (a) [2 points] Write about two benefits of MapReduce in cloud computing.
✓ (b) [2 points] What are the functions of policies in identity and access management?
✓ (c) [2 points] How does the OpenFlow protocol help in software defined networking?
- ✓ Write the boto command to
✓ (a) [1 point] launch an EC2 instance.
✓ (b) [1 point] upload a file to a S3 bucket.
✓ (c) [1 point] get IP address of Amazon EC2 instance.
✓ (d) [1 point] list running Amazon EC2 instances.
- ✓ (a) [2 points] What are the functions of publishers and subscribers in Amazon Simple Notification Service (SNS)?
✓ (b) [3 points] How do the edge locations help Amazon CloudFront to provide services?
- ✓ (a) [2 points] Can service oriented architecture (SOA) achieve interoperability? Explain.
✓ (b) [3 points] How can we create the scale up and scale down policies in Amazon Auto Scaling Group? Write the codes also.
- ✓ (a) [2 points] Explain the deployment design step of Cloud Component Model (CCM) with a diagram.
✓ (b) [3 points] Suppose you need to create a document storage application. The users of the application can upload files, view files as well as delete files. The application should also manage the user data. Explain the architecture design of this application with a diagram. How can Django framework help to build this application?